James Keller – GES408 – Hypothesis Testing - Spreadsheet

State an appropriate null and alternative hypothesis.

H_0 – The mean Tuna Weight in kilograms is not different between populations in Marine protected areas versus those not within marine protected areas.

$H_0: \mu_1 = \mu_2$

H_A – The mean Tuna Weight in kilograms is different between populations in Marine protected areas versus those not within marine protected areas.

$H_A: \mu_1 \neq \mu_2$

We randomly sampled our tuna, and we are going to assume a normal distribution for the sake of the person writing this assignment. Have we met the third assumption of our standard t-test? How do you know?

We have met the third assumption that the variance is the same between the two samples because there is less than a 3-fold difference between the two samples.

Is the test statistic that we calculated greater than or less than the corresponding critical value? Given this, should we reject our null hypothesis?

Our test statistic is greater than the corresponding critical value. Given this, we would reject our null hypothesis.

Mean A	26.15
Mean B	19.23
Standard Deviation A	5.21
Standard Deviation B	4.81
Variance A	27.15641026
Variance B	23.10192308
Sample Size A	40
Sample Size B	40
Pooled Sample Variance	25.12916667
Standard Error of the Difference Between Two Sample Means	1.120918522
Test Statistic	6.177969106
Degrees of Freedom	78
Critical Value (from table - we are doing a 2 sided test, and using a=0.05)	1.911

$$s_{diff} = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$$

$$SE_{pooled} = \sqrt{s_{diff}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$t_{calc} = \frac{\overline{Y_1} - \overline{Y_2} - (\mu_1 - \mu_2)}{SE_{pooled}}$$